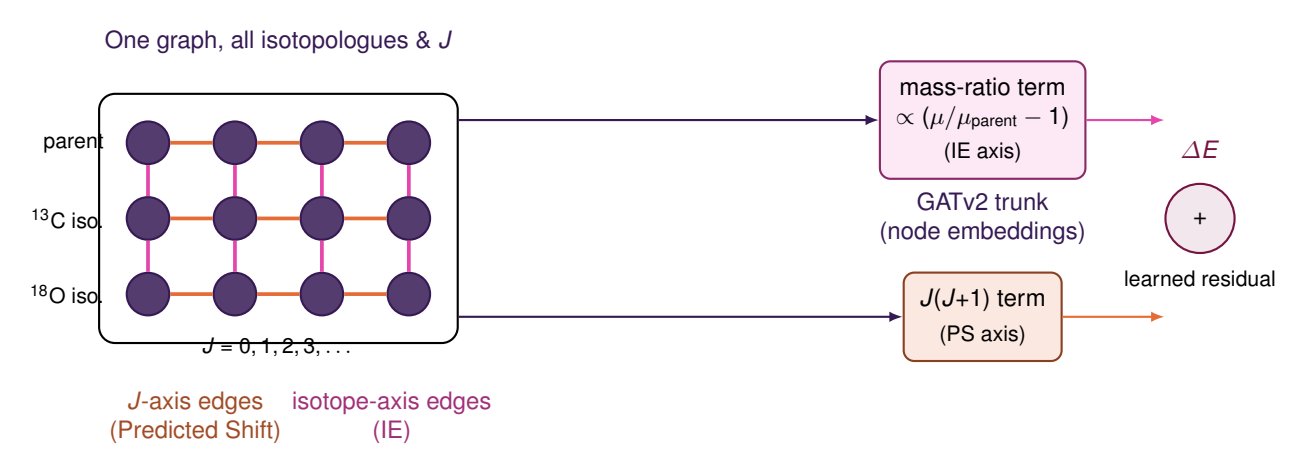
One graph, all isotopologues & J
parent
13C iso.
18O iso.
J = 0, 1, 2, 3, . . .
J-axis edges
(Predicted Shift)
isotope-axis edges
(IE)
mass-ratio term
∝ (μ/μparent − 1)
(IE axis)
GATv2 trunk
(node embeddings)
J(J+1) term
(PS axis)
ΔE
+
learned residual